MAINCRAFTS TECHNOLOGY
TASK 3 REPORT
Sequential Circuits using Verilog HDL

Sequential circuits are digital circuits whose outputs depend on the present inputs and previously stored states. This report presents the design, simulation, and verification of a D Flip-Flop, JK Flip-Flop, 4-bit Register, and 4-bit Counter using Verilog HDL in Xilinx Vivado.

D Flip-Flop

A D Flip-Flop is an edge-triggered storage element that captures the input D at every positive edge of the clock and stores it until the next active clock edge. It is widely used in registers, memories, and pipeline circuits.

Clock	D	Q(next)
↑	0	0
↑	1	1

Simulation Results: The following Vivado waveforms verify the expected behavior of the circuit.

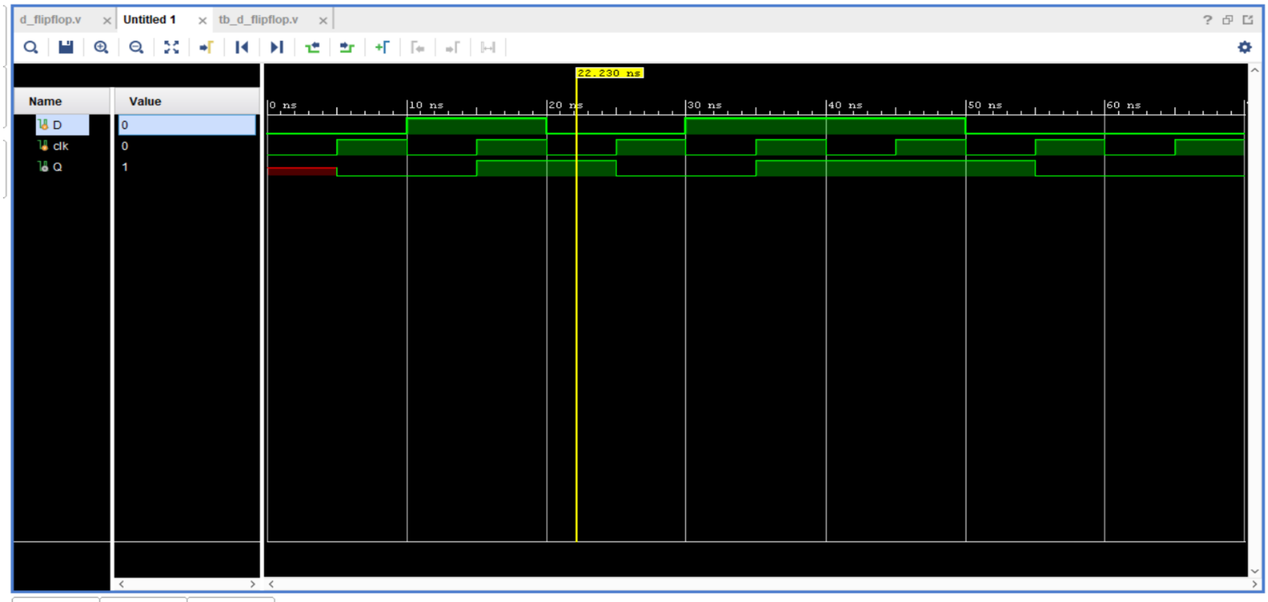

Figure 1: Functional Simulation

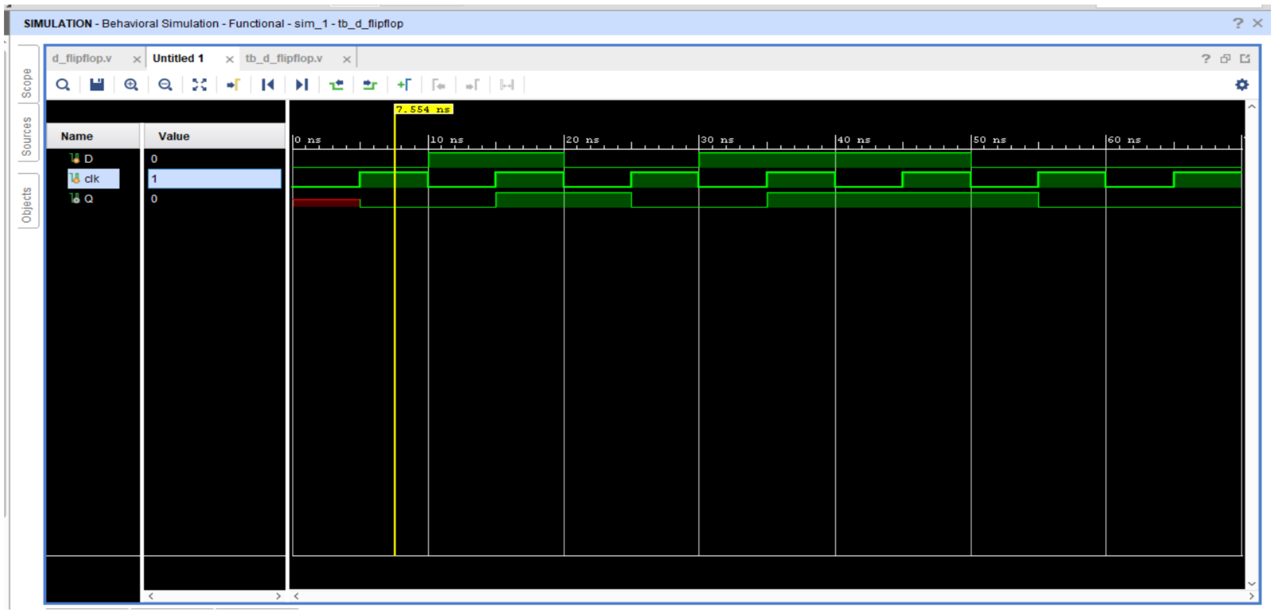

Figure 2: Functional Simulation

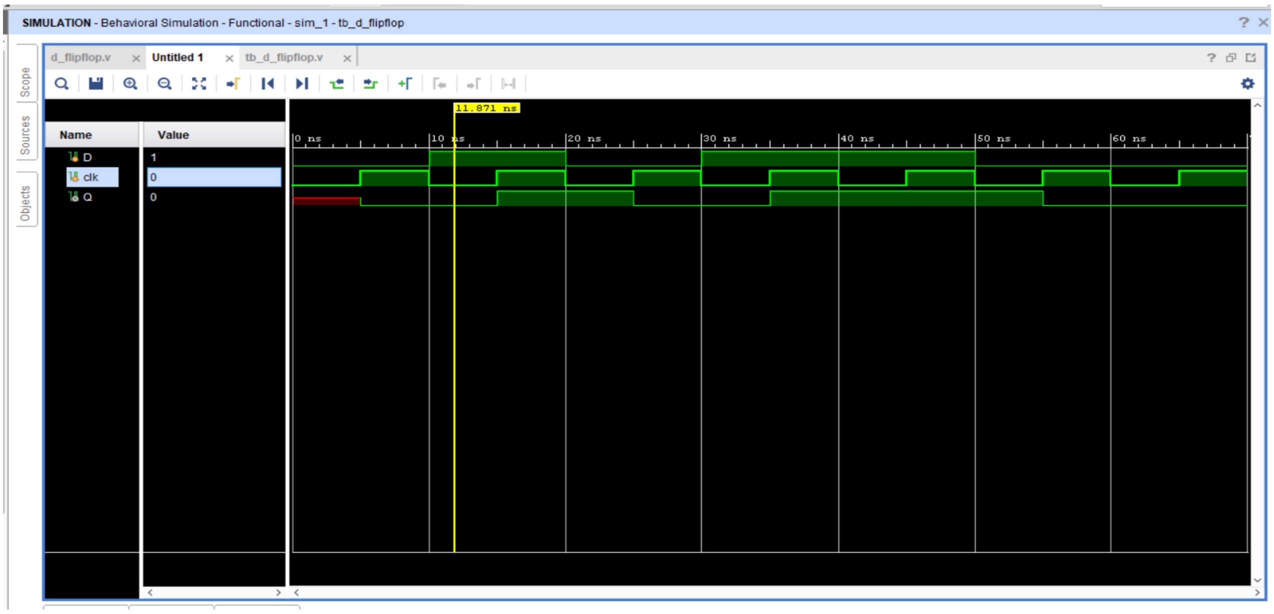

Figure 3: Functional Simulation

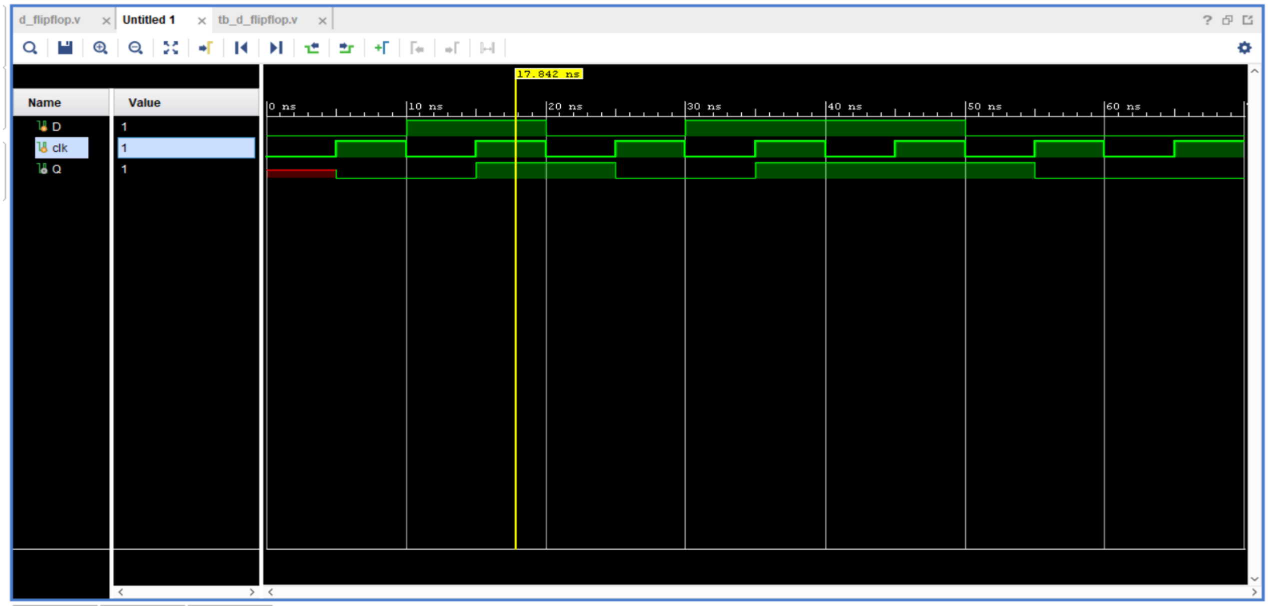

Figure 4: Functional Simulation

JK Flip-Flop

The JK Flip-Flop removes the invalid state of the SR Flip-Flop. Depending on J and K inputs it performs Hold, Reset, Set, or Toggle operation.

J	K	Operation
0	0	Hold
0	1	Reset
1	0	Set
1	1	Toggle

Simulation Results: The following Vivado waveforms verify the expected behavior of the circuit.

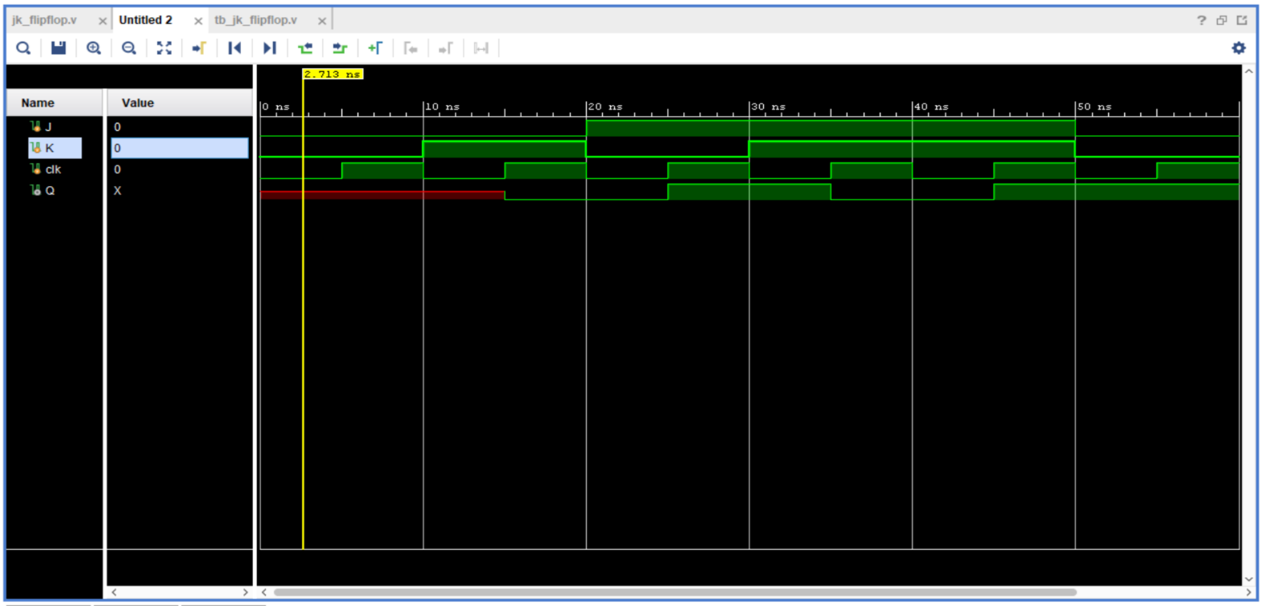

Figure 1: Functional Simulation

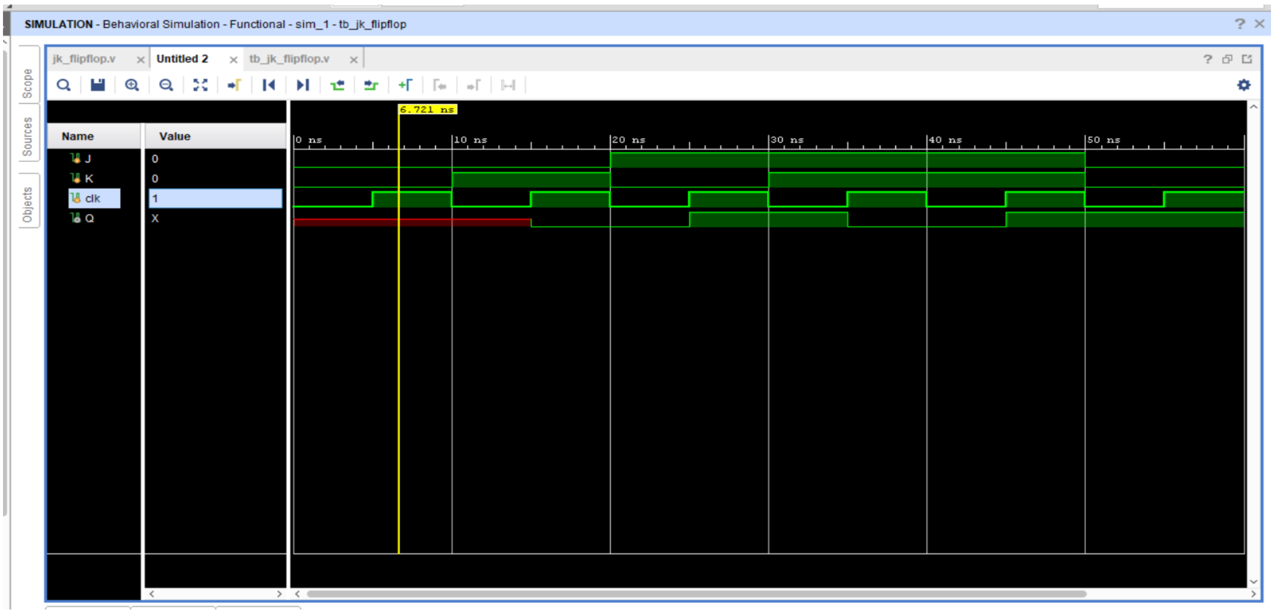

Figure 2: Functional Simulation

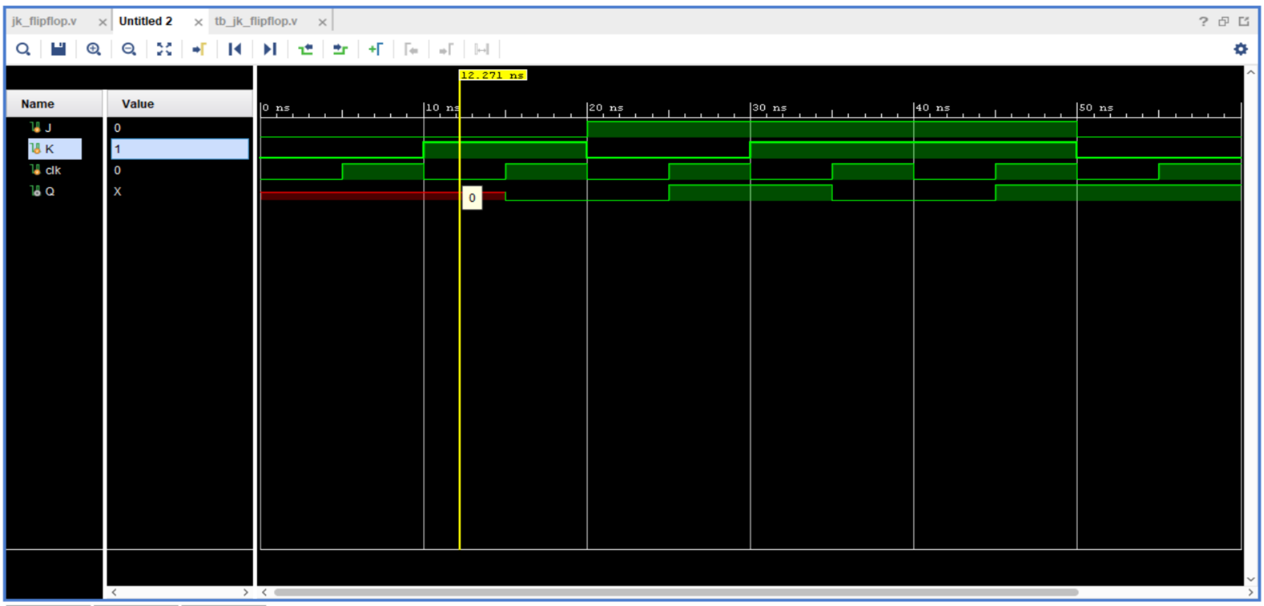

Figure 3: Functional Simulation

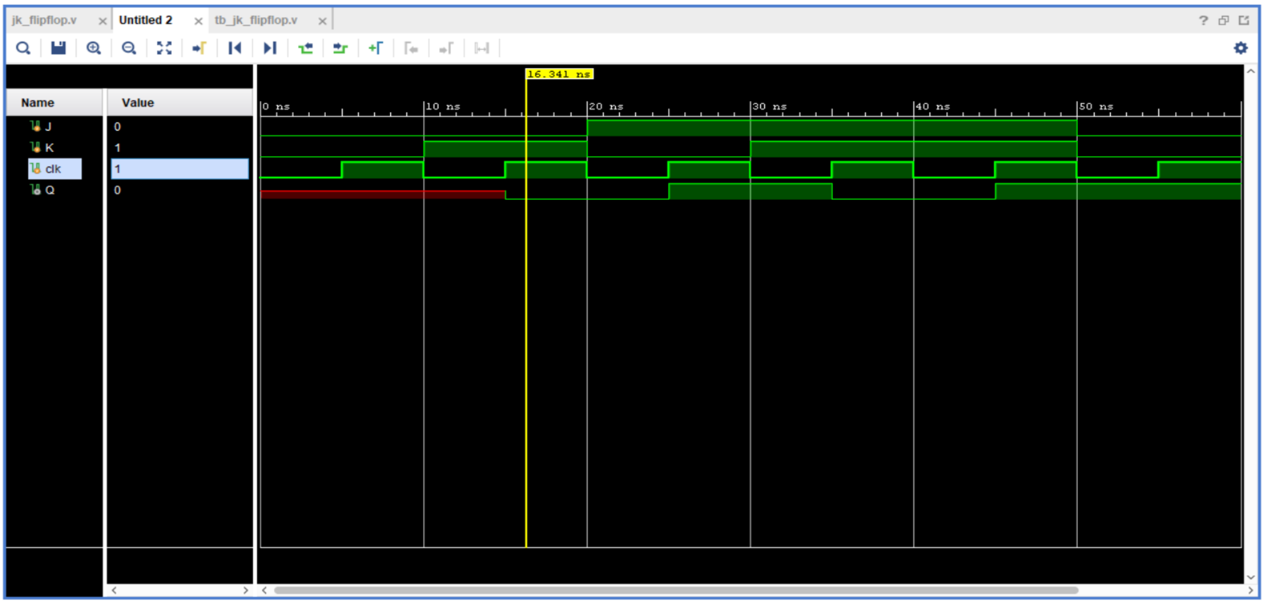

Figure 4: Functional Simulation

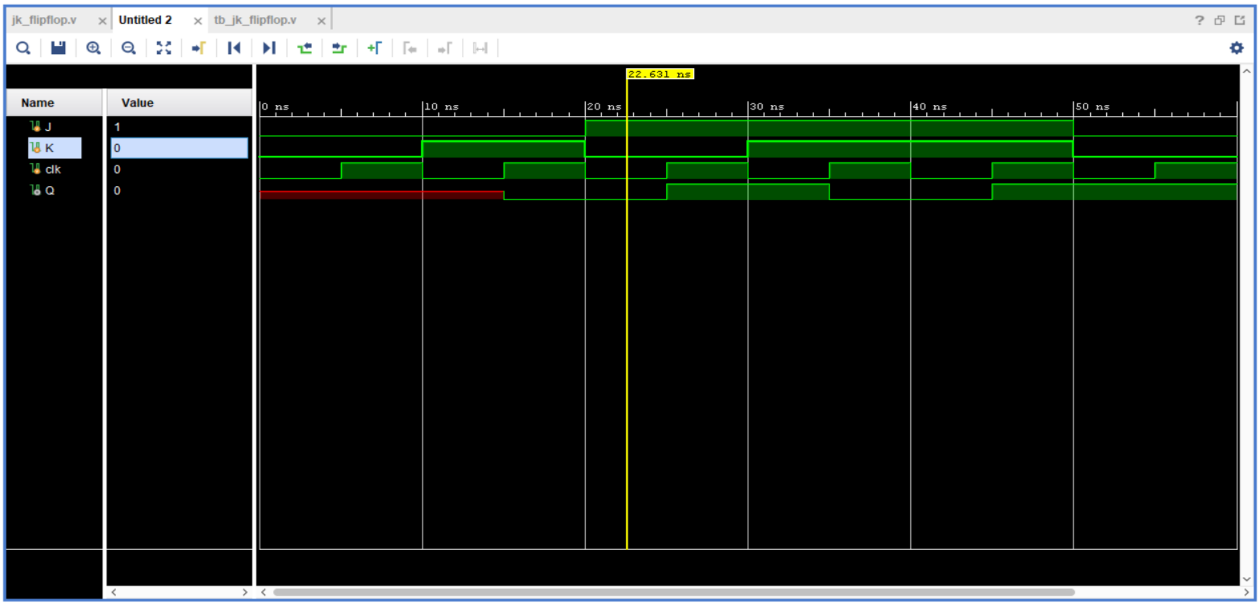

Figure 5: Functional Simulation

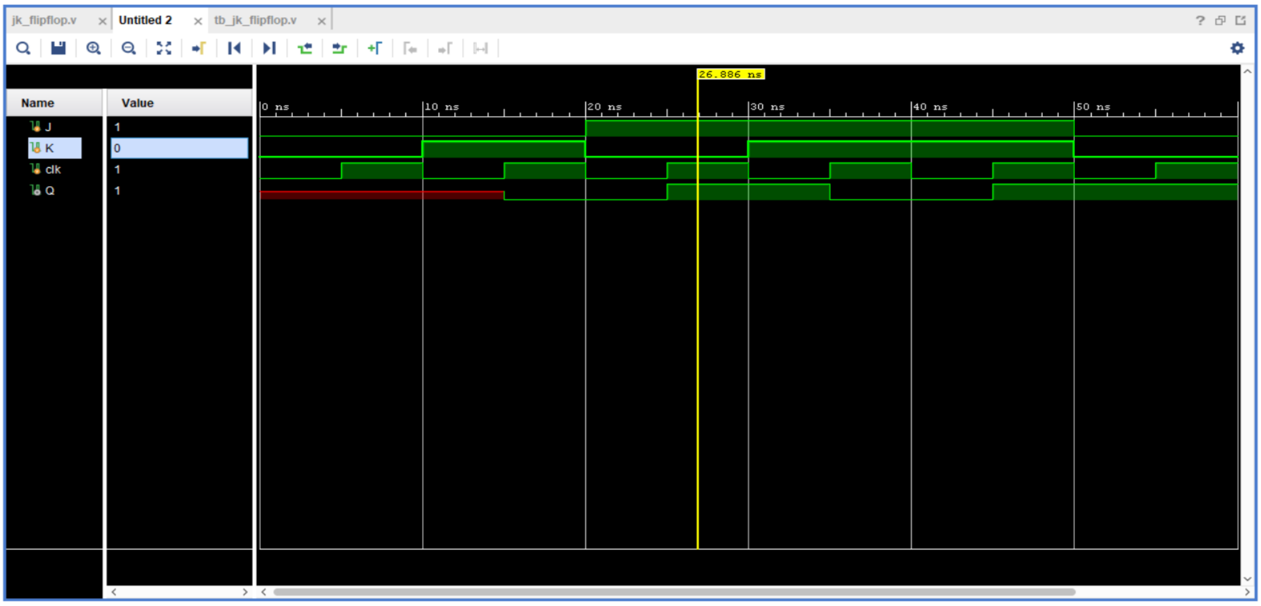

Figure 6: Functional Simulation

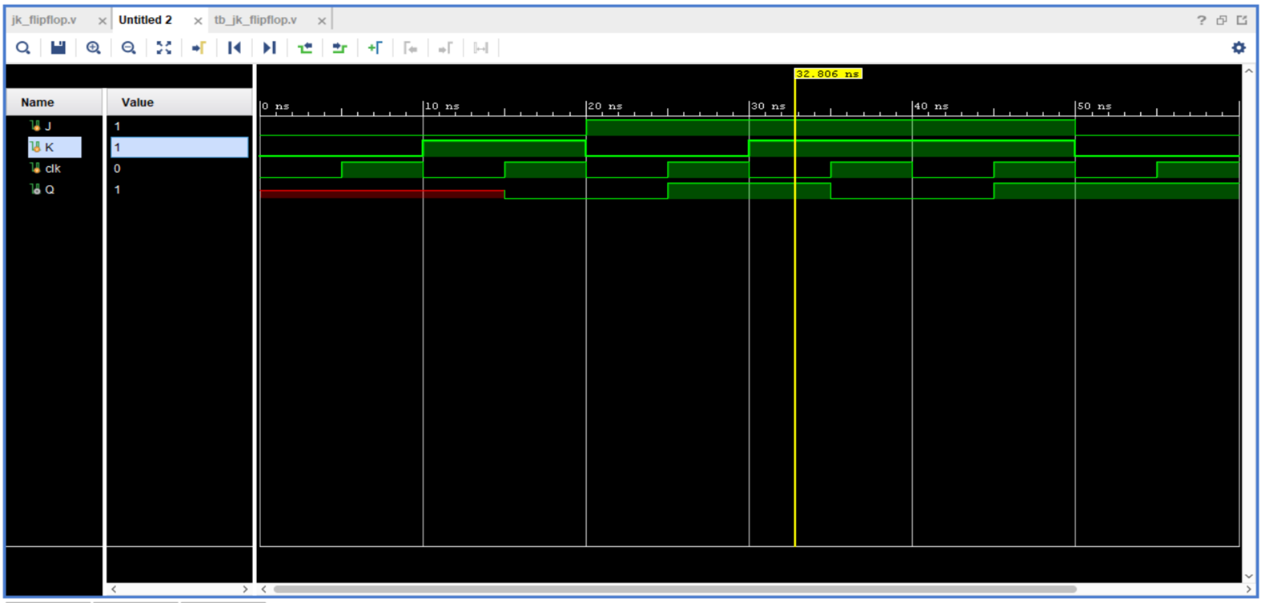

Figure 7: Functional Simulation

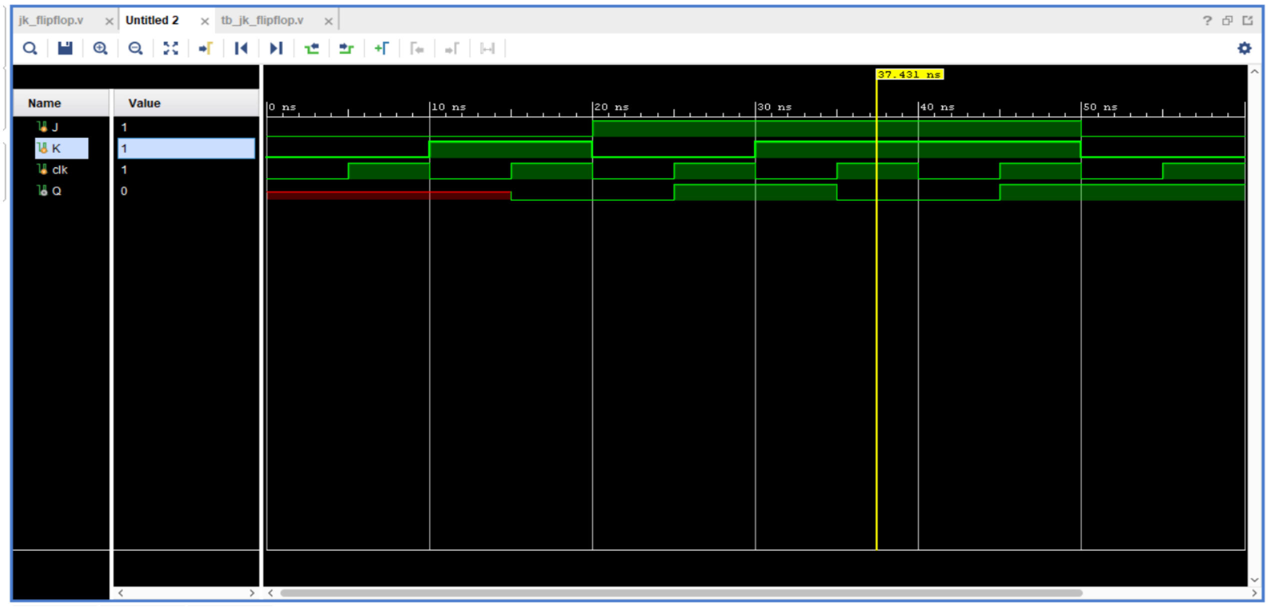

Figure 8: Functional Simulation

4-bit Register

A 4-bit register stores four bits simultaneously on each rising clock edge and holds the value until the next clock pulse.

Clock	Operation
$\uparrow$	$Q \leftarrow D$

Simulation Results: The following Vivado waveforms verify the expected behavior of the circuit.

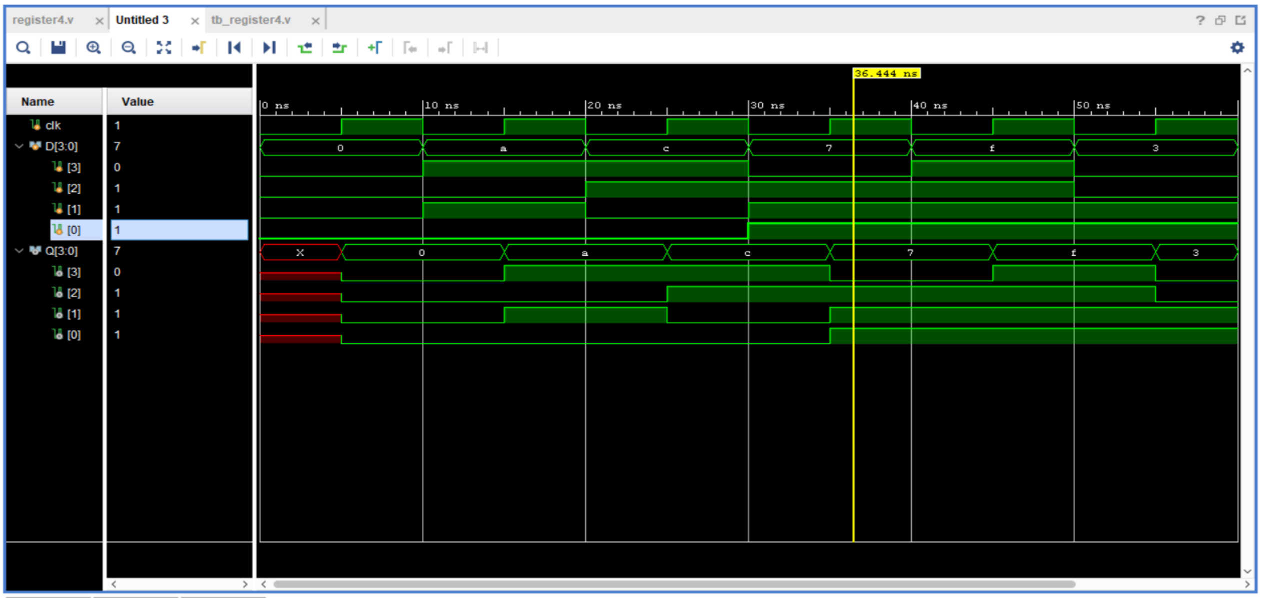

Figure 1: Functional Simulation

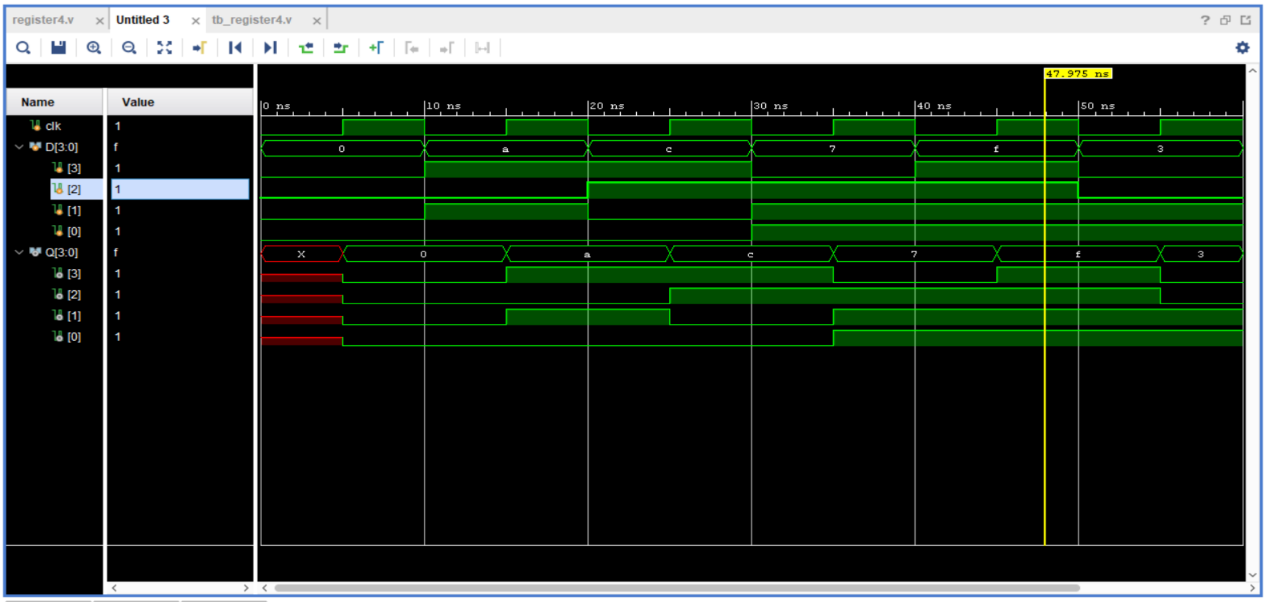

Figure 2: Functional Simulation

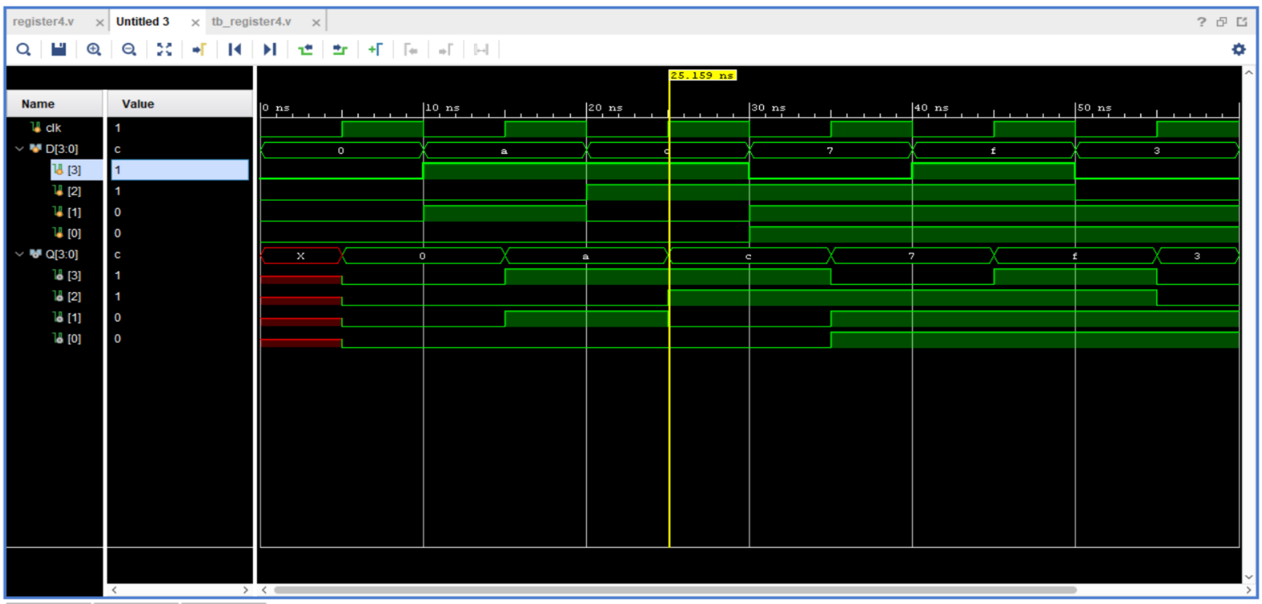

Figure 3: Functional Simulation

4-bit Counter

A 4-bit synchronous counter increments from 0000 (0) to 1111 (15) and then rolls back to 0000.

State	Binary
0	0000
1	0001
...	...
15	1111

Simulation Results: The following Vivado waveforms verify the expected behavior of the circuit.

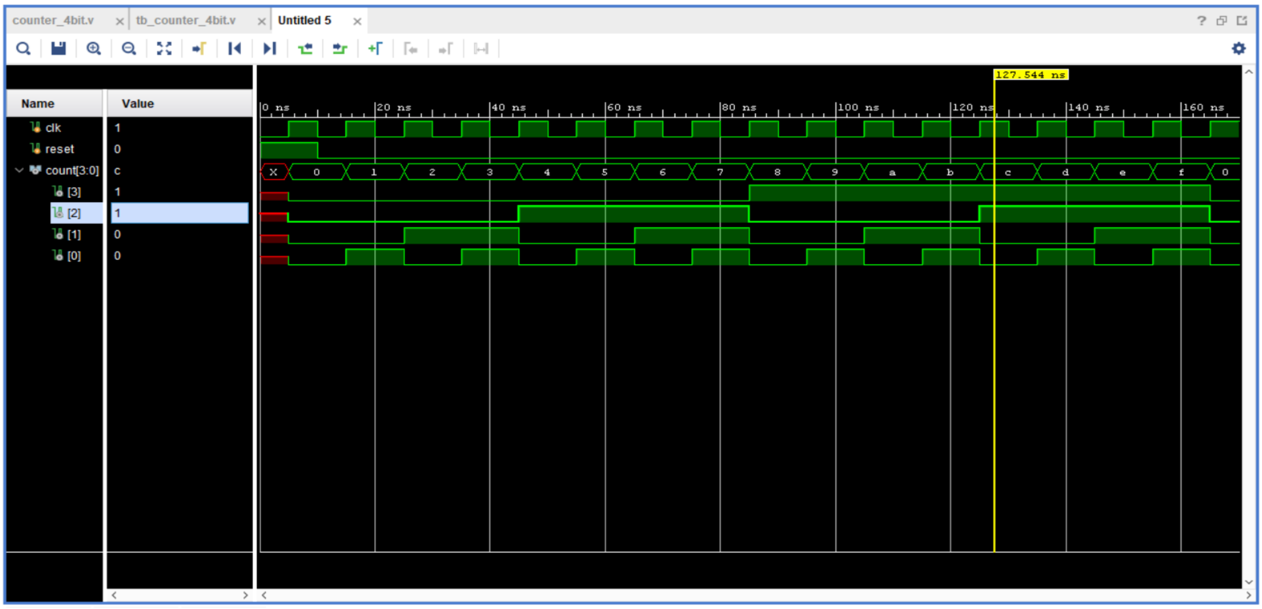

Figure 1: Functional Simulation

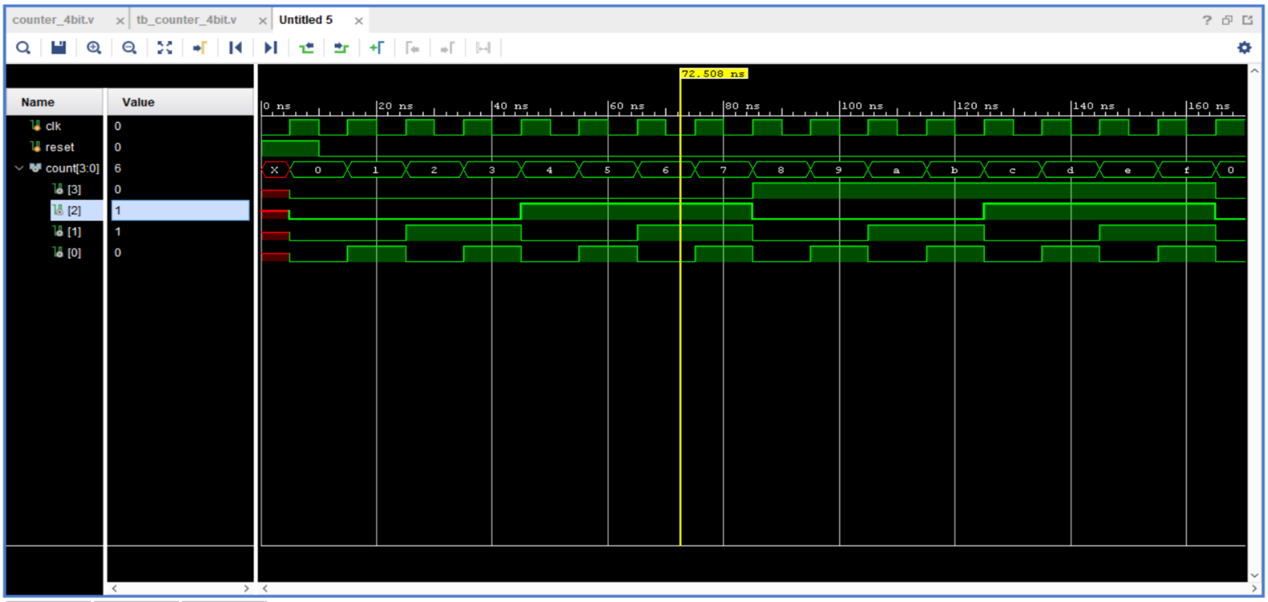

Figure 2: Functional Simulation

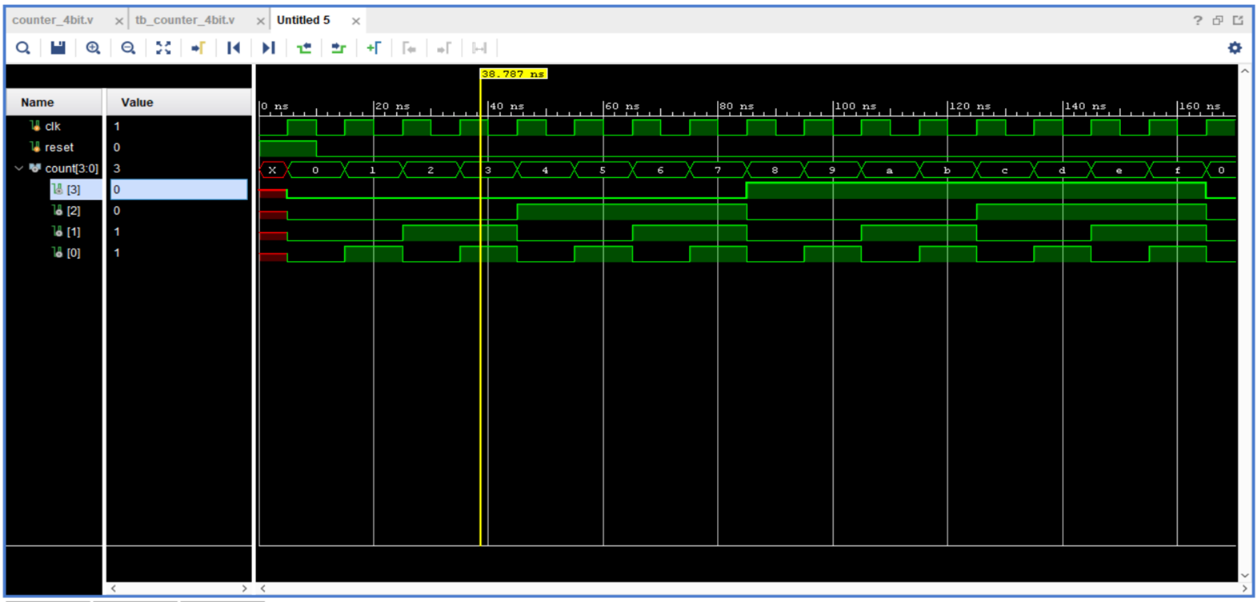

Figure 3: Functional Simulation

Conclusion

All Verilog modules compiled and simulated successfully. The D Flip-Flop stored data correctly, the JK Flip-Flop performed Hold, Reset, Set, and Toggle operations, the 4-bit Register loaded parallel data correctly, and the 4-bit Counter counted from 0 to F before rollover. The obtained waveforms matched the theoretical behavior.